

Forensics & Steganography Tools

The forensics stack in Kali

Tool	Purpose
Autopsy / Sleuth Kit	Disk imaging + artifact analysis GUI
Volatility 3	Memory forensics (processes, network, injected code)
Binwalk	Firmware/archive extraction + file signature analysis
Foremost / Scalpel / Bulk Extractor	Carve deleted/recovered files by signature
testdisk / photorec	Partition recovery + file recovery

Carve a file

```
# Extract files that match signatures from a disk image
foremost -i evidence.dd -o carved/
scalpel -i evidence.dd -o carved2/
bulk_extractor evidence.dd -o be_out/
```

Memory analysis with Volatility 3 (Python)

```
# find running processes and args
vol3 -f mem.dump windows.pslist
vol3 -f mem.dump windows.pstree
vol3 -f mem.dump windows.netstat
vol3 -f mem.dump windows.malfind
# dump a process's memory
vol3 -f mem.dump windows.dumpfiles --pid 1234
```

malfind flags suspect injected code – your hint that a process hosts shellcode.

Steganography: hide & seek

```
# hide a message file inside an image (steghide)
steghide embed -cf photo.jpg -sf secret.txt -p 'passphrase'
steghide extract -sf photo.jpg -p 'passphrase'

# file-inspecting: what else is inside?
binwalk secret.jpg
strings secret.jpg | grep -i 'flag|secret'
```

Real forensic mindset

Preserve first: image, hash, work on the copy

Timelines beat single artifacts: build the story

Stego is common in example challenges and real malware delivery
(payload hidden in image)

Kali's tools answer the question "what is on this media?" – a crime scene needs procedure too

Combine: a phishing email + stego image + memory dump = a realistic ransomware-intrusion lab the whole defense module will revisit.

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